Statistical Inferences about Parameters

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‘Fitting’ Parameters

- As we have discussed, the sum of squared errors is an excellent method for measuring total error, for reasons of efficiency, mathematical practicality, and tradition.

- This also means (and we have shown) that the sample mean is an unbiased but not necessarily accurate estimate of the population mean.

- Armed with this knowledge, we can use the tools we already have to answer some important questions.

Does having the flu increase your temperature?

- Let's say that I have a hypothesis that having the flu increases your body's temperature.
- I formulated this hypothesis based on the anecdotal observation that when I touch the foreheads of sick people, they seem warmer than me.
- I randomly sample ten people with the flu and take their surface temperature on the forehead. (Arguably not the best, but at least a non-invasive measure of temperature.)

Does having the flu increase your temperature?

37.57	**37.98**
37.79	**38.08**
36.74	**37.97**
37.63	**38.26**
37.98	**36.59**

- What is the sum of squared errors for 37*C?

Does having the flu increase your temperature?

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36.74	37.97
37.63	38.26
37.98	36.59

0.57	0.98
0.79	1.08
-0.26	0.97
0.63	1.26
0.98	-0.41

- What is the sum of squared errors for 37*C?

- 7.197

Does having the flu increase your temperature?

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- What is the sum of squared errors if we “fit” beta with our sample mean?

- Sample mean = 37.659

Does having the flu increase your temperature?

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37.63	38.26
37.98	36.59

-0.089	0.321
0.131	0.421
-0.919	0.311
-0.029	0.601
0.321	-1.069

- What is the sum of squared errors if we "fit" beta with our sample mean?
- Sample mean = 37.659
- 2.854

Does having the flu increase your temperature?

Residuals 1 **SSE(C) = 7.197**

0.57	0.98
0.79	1.08
-0.26	0.97
0.63	1.26
0.98	-0.41

Residuals 2 **SSE(A) = 2.854**

-0.089	0.321
0.131	0.421
-0.919	0.311
-0.029	0.601
0.321	-1.069

- Which estimate is a better explanation of our data?
- To answer this question, we can calculate the proportional reduction in error, ***PRE***.
- $PRE = \frac{SSE(C) - SSE(A)}{SSE(C)}$

Does having the flu increase your temperature?

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- Which estimate is a better explanation of our data?
- To answer this question, we can calculate the proportional reduction in error, ***PRE***.
- $PRE = \frac{7.197-2.854}{7.197} = 0.603$

Does having the flu increase your temperature?

- Thus, using our sample mean led to ~60% less error than estimating the average human body temperature of 37*C.

- But does this mean that having the flu increases your temperature? How much better does an explanation need to be before we decide it is the better explanation?

- How can we use the proportional reduction in error to make an ***inference*** from our sample to the population?

The sampling distribution of PRE

- Recall that the mean will always produce the smallest SSE for a given set of numbers. Thus, it is not unusual that SSE(A) < SSE(C).

- In order to make an inference, therefore, we need to see if SSE(A) is sufficiently smaller than SSE(C) to make the result ***statistically interesting***.
  - That is, is the reduction in SSE large enough that we think it is ***unlikely*** to happen by ***sampling variability alone***?

The sampling distribution of PRE

- In class, we will show the sampling distribution of PRE using simulations.

- For now, let's focus on two key concepts:
  - The proportional reduction in error will always be ≥ 0 in this situation, because the sample mean will provide the smallest SSE for a set of numbers. (If the mean had been 37 exactly, PRE would be 0.)

  - The larger the proportional reduction in error, the less likely we think a result is to have occurred from sampling variability alone.
    - (That is, if the variability explained by the estimate is large compared to the residual variability, there might be something ***systematic*** rather than ***random*** happening in our data.)

The sampling distribution of PRE

- "Know thy distributions."

- In order for us to make an inferential statement about the probability of observing PRE, we need to make an estimate about the variance in the population.

- As we have shown, the sample variance is an unbiased estimate of the population variance:
  - $\sigma^2 \approx s^2 = \frac{SSE}{(n-1)} = \frac{\sum(y_i - \bar{y})^2}{(n-1)}$

The sampling distribution of PRE

- Critically, the shape of the sampling distribution depends on the number of estimated parameters in Model A (PA) and the number of observations in the sample (n).
  - $\sigma^2 \approx s^2 = \frac{SSE}{(n-PA)} = \frac{\sum(y_i-\bar{y})^2}{(n-1)}$

- We are estimating one parameter (the mean) and we have 10 observations.
  - As the number of observations increases, s^2 will be a more consistent approximation of σ^2 and thus a larger PRE is more surprising.
  - As the number of parameters increases, we expect to explain more error (even just by chance) and thus a larger PRE is less surprising.

The sampling distribution of PRE

- Thus, the number of parameters in Model A (PA) and the number of observations in the sample (n) work against each other in how we interpret the PRE we observe.

- However, critical values of PRE and its distribution are rare in statistics books. Instead, it is much more common to see PRE converted into an ***F-value***, and interpreted with respect to the ***F-distribution***.